

Relatum: Autonomous Mathematical Discovery via Relational Closure and Property Implication

Abstract

We present Relatum, a relation-first reasoning engine that autonomously discovers algebraic structure through two independent mechanisms: *inductive closure scoring* and *property implication analysis*. Starting from an exhaustive enumeration of all 19,683 binary operations on a 3-element carrier, the closure-scoring mechanism partitions the model space into 10 axiom classes and identifies the abelian group region as optimal through dual-signal alignment of model-space rarity and closure-space richness. A formal equivalence proof establishes that the discovered rule bundle is logically equivalent to the group axiom subset $\{G_2, G_3\}$, holding for all groups including infinite ones.

Independently, the property implication mechanism defines algebraic properties (left identity, right identity, idempotent) as first-class relational objects and discovers their pairwise equivalence or strict implication by comparing property extensions. Exhaustive verification on all 935 non-trivial 3-element models finds: (1) identity presence is both necessary and sufficient for left-right identity equivalence (verified on 935/935 models); (2) inverse presence guarantees the identity as the unique idempotent, but commutativity with identity alone yields this equivalence in only 75% of models, revealing structural diversity within axiom classes that single-representative sampling cannot detect; (3) associativity is independent of both equivalences.

The two mechanisms use entirely different inference paths—closure generativity scoring versus property extension inclusion—yet produce consistent structural classifications. This cross-validation, obtained without human guidance on what to look for, constitutes the central result: the algebraic boundary between groups and weaker structures is discoverable from raw operation tables by multiple independent methods within a single relational framework.

Beyond analysis, the system constructs new properties via conjunction and negation, scored by a combination of independence, rarity, and constraint. On a 4-element chain, it autonomously discovers the concept of “interior elements” ($\neg\text{maximal} \wedge \neg\text{minimal}$) — a recognized mathematical concept found without being told its name. A cross-degenerate filter eliminates combinations that merely reconstruct existing properties, reducing the false discovery rate from 67% to 0%.

The engine also supports pattern-level deduction (variable-to-variable unification, stratified negation, ω -rule inductive promotion, and proof by contradiction), demonstrated on ZFC set-theoretic axioms where the system derives universally quantified conclusions such as $\forall A. \emptyset \in \mathcal{P}(A)$ from two input propositions in 0.00s with 11 total facts.

1 Introduction

A central question in automated mathematical reasoning is whether a single computational framework can both *discover* mathematical structure from raw data and *verify* that the discovered structure reflects genuine mathematical truth rather than computational artifact. Most existing systems address one side: automated theorem provers verify conjectures but do not discover them; machine learning systems discover patterns but cannot certify their mathematical status.

We present Relatum, a relation-first reasoning engine that addresses both sides within a unified relational closure framework. The system operates on a minimal ontology—terms, relations, and inference rules—with no built-in mathematical semantics. Algebraic properties, set-theoretic

axioms, and order-theoretic concepts are all expressed as relational declarations, and all reasoning reduces to closure computation.

1.1 Contributions

This paper makes five contributions:

1. **Inductive algebraic discovery** (Section 3). Starting from all 19,683 binary operations on a 3-element carrier, the system autonomously partitions the model space into 10 axiom classes and identifies the abelian group as the optimal region through dual-signal alignment of model-space rarity and closure-space richness. A formal equivalence proof establishes that the discovered rules are logically equivalent to group axioms.
2. **Property implication analysis** (Section 4). Properties are defined as first-class relational objects. The system discovers pairwise implication and equivalence between properties by comparing their extensions. Across 10 axiom classes, it finds that identity presence determines left-right identity equivalence, and that associativity is independent of this equivalence.
3. **Cross-domain generality** (Section 5). The property implication mechanism is validated on three unrelated domains—algebra (10 axiom classes), order theory (241 finite posets), and set theory (ZFC powerset chain). Four structural meta-patterns repeat across all three domains, establishing the mechanism as domain-general.
4. **Failure mode diagnosis** (Section 5). When property equivalences fail to be confirmed, the system distinguishes *structural failure* (genuine mathematical non-equivalence, constant across resource levels) from *resource failure* (definitionally true equivalence that cannot be inductively confirmed due to derivation-rate competition under a finite fact budget).
5. **Property composition and concept discovery** (Section 6). The system constructs new properties via conjunction and negation, scored by independence, rarity, and constraint. On a 4-element chain, it autonomously discovers “interior elements” ($\neg$ maximal $\wedge$ $\neg$ minimal). A cross-degenerate filter eliminates combinations that reconstruct existing properties.

1.2 System capabilities

The engine supports six reasoning mechanisms, each introduced as a relational extension rather than a separate subsystem:

- **Pattern facts**: variable-to-variable unification, enabling universally quantified conclusions (Section 2).
- **Stratified negation**: derive properties defined by absence (e.g., maximal elements).
- **ω -rule**: inductive promotion of saturated ground chains to pattern facts.
- **Proof by contradiction**: hypothetical reasoning via engine cloning.
- **Property implication**: first-class properties with observed implication and equivalence.
- **Proto-level axiom discovery**: beam search over relational composition candidates with compression + purity scoring, demonstrated by autonomous discovery of transitivity from raw binary relation data (Section 3).

All mechanisms share the same closure engine, the same fact representation, and the same rule-matching infrastructure.

2 System Architecture

Relatum operates on three primitives: *terms*, *relations*, and *rules*. All mathematical content is encoded in these primitives; the engine has no built-in notion of equality, ordering, or algebraic operation.

2.1 Terms

A term is either a variable $?x$, a constant a , or a function application $f(t_1, \dots, t_n)$. Constants are nullary applications. Compound terms serve as Skolem witnesses (e.g., $power(empty)$) and are bounded by a configurable depth limit (default 8).

2.2 Relations and Facts

A relation $R(t_1, \dots, t_n)$ is a named tuple of terms. A *ground fact* contains no variables. A *pattern fact* contains free variables and represents a universally quantified assertion: $R(empty, _0)$ means $\forall x. R(empty, x)$.

Pattern facts participate in rule matching via bidirectional unification. When a rule premise $R(?s, ?a)$ matches a pattern fact $R(empty, _0)$, the substitution $\{s \mapsto empty, a \mapsto _0\}$ propagates the free variable into the conclusion, producing new pattern facts without ground instantiation.

Concrete example. Starting from one ground fact $set(empty)$ and the pattern axiom $subset(empty, _0)$ (“empty is a subset of any set”), the engine derives:

1. The rule $subset(?s, ?a), set(?s) \vdash member(?s, power(?a))$ matches:
 - premise 1 against $subset(empty, _0)$: $\{s \mapsto empty, a \mapsto _0\}$ (variable-to-variable binding),
 - premise 2 against $set(empty)$: $s = empty$ confirmed.
2. Conclusion: $member(empty, power(_0))$ — a universally quantified pattern fact asserting $\forall A. \emptyset \in \mathcal{P}(A)$.

This derivation completes in 0.00s with 11 total facts, producing the universal theorem from one ground fact and one pattern axiom. No enumeration of specific sets is needed.

2.3 Closure Engine

The engine computes the least fixed point of a set of inference rules applied to an initial fact set. Each round applies:

1. **User rules** (stratum 0): positive premises matched against the fact set.
2. **Reflexivity**: $R(t, t)$ for every ground term t and every reflexive relation R .
3. **Congruence**: structural replacement under equivalence relations.
4. **Axiom instantiation**: ground substitution of universally quantified equations.

After stratum 0 reaches a fixed point, higher strata execute in order:

- **Stratum 1+: Stratified negation.** Rules with negated premises fire only when the negated pattern is *absent* from the completed fact set. Multiple negation strata are supported; if a higher stratum produces new facts, the engine re-runs from stratum 0 (iterative stratification).

- **Refutation phase.** Rules declared with `with_refutation` perform proof by contradiction: for each binding from positive premises, the engine clones itself, hypothetically adds facts, runs closure, and checks for a contradiction relation. If all hypothetical branches contradict, the conclusion is established.
- **ω -rule.** After saturation, the engine scans for unary relations with inductive structure (base case + step rule $R(x) \vdash R(f(x))$). If found, it promotes $R(_0)$ as a pattern fact and re-runs closure. A defense mechanism blocks non-ground conclusions from pure-transfer rules ($R(x) \vdash S(x)$) to prevent incorrect universalization.
- **Property implication.** If properties are defined, the engine scans all unary property extensions. When $\text{ext}(P) \subseteq \text{ext}(Q)$ with $\text{ext}(P) \neq \emptyset$, it derives *implies_observed*(P, Q). Bidirectional inclusion yields *equivalent_observed*(P, Q). These are *observed* (inductive) implications, distinct from deductive implications, and do not trigger modus ponens.

2.4 Properties as First-Class Objects

A property is a named constant term representing a predicate. The declaration

```
define_property("left_id", &["e"],
  vec![op(e, x, x)])
```

automatically generates a detection rule $op(e, x, x) \vdash has_property_1(e, left_id)$ and marks *left_id* with *is_property*(*left_id*).

For algebraic properties requiring universal quantification (e.g., “*e* is a left identity for *all* *x*”), detection uses double negation:

$$\begin{aligned} not_left_id(e) &\leftarrow element(e), element(x), \neg op(e, x, x) \\ has_property_1(e, left_id) &\leftarrow element(e), \neg not_left_id(e) \end{aligned}$$

This correctly detects that *e* satisfies $op(e, x, x)$ for *all* elements *x*, not just some.

Properties are first-class: they can be bound to variables (*has_property_1*(?*x*, ?*p*) matches any property), quantified over, and appear in further rules.

3 Inductive Discovery via Closure Scoring

3.1 Model Space Enumeration

We enumerate all $3^9 = 19,683$ binary operations on a 3-element carrier $\{e_0, e_1, e_2\}$. Each operation is classified by which standard axioms it satisfies: associativity, commutativity, identity, and inverse. This yields 10 non-trivial axiom classes (plus a large “none” class of 18,748 operations satisfying no standard axioms).

3.2 Dual-Signal Scoring

For each axiom class, the system computes closure under candidate rules discovered by beam search. Two signals are combined:

- **Model-space rarity:** $1/\ln(\text{model count} + 1)$. Rarer classes carry more information.
- **Closure-space richness:** number of derived facts, weighted by structural diversity.

The `assoc+comm+id+inv` class (3 models—the abelian groups) scores highest: maximally rare in model space and maximally rich in closure space. This dual alignment identifies the group region as optimal without any prior mathematical knowledge.

Table 1: Axiom classes on 3-element carriers, sorted by model count.

Axiom class	Models
assoc+comm+id+inv	3
assoc+id	6
id+inv	18
comm+id	24
assoc+comm+id	24
comm+id+inv	30
assoc+comm	36
assoc	44
id	138
comm	612

3.3 Discovered Rules and Formal Equivalence

The beam search discovers a rule bundle $\{D_1, D_2, D_3\}$ that produces the richest closure on group-class models. A formal proof establishes:

Theorem 1. *The discovered rule bundle $\{D_1, D_2, D_3\}$ is logically equivalent to the group axiom subset $\{G_2, G_3\}$ (identity and inverse laws), holding for all groups including infinite ones.*

The full proof is provided in the supplementary material. The key insight is that D_1 and D_2 encode left/right identity behavior, while D_3 encodes inverse existence—precisely the group axioms beyond associativity and closure.

3.4 Transfer Verification

The discovered rules are transferred to a held-out $\mathbb{Z}_4$ structure (cyclic group of order 4, not seen during discovery). All 6 predictions are correct, confirming that the rules generalize beyond the training carrier size.

3.5 Proto-Level Axiom Discovery

The same beam search mechanism operates at the *axiom schema* level, not just the model level. Given raw binary relation data (edges $R(a, b), R(b, c), R(c, d), R(d, e)$) with no rules and no prior knowledge, the system generates 18 relational composition candidates and scores them by generativity.

The top-scoring candidate is $R(x, y), R(y, z) \vdash R(x, z)$ —transitivity—with a score of 10.10 and 11 derived facts. The system discovers transitivity as the most productive axiom schema for this binary relation, without being told what transitivity is.

When multiple relations are present simultaneously (a strict order, a symmetric relation, and a cyclic relation), a scoring function combining *compression* (how many existing facts does the axiom re-derive?) and *purity* (does the axiom stay within a single relation?) correctly identifies per-relation axioms: symmetry + transitivity for the symmetric relation, transitivity for the order, and cyclic closure for the cycle. Without the compression + purity loss, cross-relation mappings dominate; with it, each relation’s natural axioms emerge independently.

4 Property Implication Analysis

4.1 Mechanism

Given a set of named properties and a closed fact set, the system computes the *extension* of each property—the set of ground elements satisfying it—and checks pairwise inclusion:

- $\text{ext}(P) \subseteq \text{ext}(Q)$ with $\text{ext}(P) \neq \emptyset \Rightarrow \text{implies_observed}(P, Q)$
- Bidirectional inclusion $\Rightarrow \text{equivalent_observed}(P, Q)$

At fixpoint, the closure is complete: all derivable *has_property* facts have been derived. Extension inclusion is therefore a theorem of the structure, not a statistical observation. The engine promotes *implies_observed* to *implies*, enabling modus ponens to derive new property instances:

$$\text{implies}(P, Q), \text{has_property}(x, P) \vdash \text{has_property}(x, Q)$$

This can extend property extensions beyond what direct detection rules produce, breaking through the original fixpoint.

4.2 Algebraic Properties

On each of the 10 axiom classes (Table 1), we define three properties via universal (double-negation) detection:

- *left_id*(*e*): $\forall x. \text{op}(e, x, x)$
- *right_id*(*e*): $\forall x. \text{op}(x, e, x)$
- *idempotent*(*e*): $\text{op}(e, e, e)$

4.3 Single-Representative Analysis

Selecting one representative operation table per axiom class yields the initial property equivalence table:

Table 2: Property equivalences from single-representative sampling.

Axiom class	left $\leftrightarrow$ right	left $\leftrightarrow$ idem	right $\leftrightarrow$ idem
assoc	×	×	×
assoc+comm	×	×	×
assoc+comm+id	$\leftrightarrow$	×	×
assoc+comm+id+inv	$\leftrightarrow$	$\leftrightarrow$	$\leftrightarrow$
assoc+id	$\leftrightarrow$	×	×
comm	×	×	×
comm+id	$\leftrightarrow$	$\leftrightarrow$	$\leftrightarrow$
comm+id+inv	$\leftrightarrow$	$\leftrightarrow$	$\leftrightarrow$
id	$\leftrightarrow$	×	×
id+inv	$\leftrightarrow$	$\leftrightarrow$	$\leftrightarrow$

From this table, three structural laws appear to hold:

1. Identity presence determines left–right equivalence.
2. Inverse (or commutativity with identity) determines identity–idempotent equivalence.

3. Associativity is independent of both equivalences.

Finding (1) and (3) survive exhaustive verification (Section 4.4). Finding (2) does not.

4.4 Exhaustive Verification

To move beyond single-representative sampling, we run property implication on *every* model in each axiom class—935 non-trivial models total. All closures run to saturation with Phase 9 promotion (implies $\rightarrow$ modus ponens).

Table 3: Exhaustive property equivalence across all 935 models. $\checkmark$ = holds on all models (complete verification). Fractions indicate partial equivalence within the class.

Axiom class	Models	left $\leftrightarrow$ right	left $\leftrightarrow$ idem	right $\leftrightarrow$ idem
assoc	44	$\times$ (0/44)	1/44	1/44
assoc+comm	36	$\times$ (0/36)	$\times$ (0/36)	$\times$ (0/36)
assoc+comm+id	24	$\checkmark$ (24/24)	$\times$ (0/24)	$\times$ (0/24)
assoc+comm+id+inv	3	$\checkmark$ (3/3)	$\checkmark$ (3/3)	$\checkmark$ (3/3)
assoc+id	6	$\checkmark$ (6/6)	$\times$ (0/6)	$\times$ (0/6)
comm	612	$\times$ (0/612)	$\times$ (0/612)	$\times$ (0/612)
comm+id	24	$\checkmark$ (24/24)	18/24	18/24
comm+id+inv	30	$\checkmark$ (30/30)	15/30	15/30
id	138	$\checkmark$ (138/138)	54/138	54/138
id+inv	18	$\checkmark$ (18/18)	$\checkmark$ (18/18)	$\checkmark$ (18/18)

4.5 Single-Representative vs. Exhaustive

The left–right column is perfectly binary: $\checkmark$ whenever **id** is present, $\times$ otherwise. Single-representative sampling and exhaustive verification agree completely. Finding (1) is confirmed as a universal law on 3-element carriers.

The left–idem column tells a different story. Single-representative sampling reported $\checkmark$ for **comm+id**, **comm+id+inv**, and **id**. Exhaustive verification reveals these are partial:

- **comm+id**: 18/24 (75%), not 24/24.
- **comm+id+inv**: 15/30 (50%), not 30/30.
- **id**: 54/138 (39%), not 138/138.

Only **id+inv** (18/18) and **assoc+comm+id+inv** (3/3) show true universal equivalence between identity and idempotency. Finding (2) from Section 4.3 was an artifact of single-representative bias.

4.6 The Algebraic Source of Diversity

The 25% of **comm+id** models where left–idem fails share a concrete algebraic feature: *extra idempotents*.

Concrete example. Model #5 in the **comm+id** class has operation table:

$$\begin{pmatrix} 0 & 1 & 2 \\ 1 & 1 & 2 \\ 2 & 2 & 0 \end{pmatrix}$$

Here $e_0 = 0$ is the identity element ($0 \cdot x = x$ for all x), and $left_id = \{e_0\}$. But e_1 is also idempotent ($1 \cdot 1 = 1$) while not being an identity ($1 \cdot 0 = 1 \neq 0$). So $idempotent = \{e_0, e_1\} \supsetneq \{e_0\} = left_id$.

In algebraic terms, e_1 acts as a *left zero* on a substructure: it absorbs some inputs without being an identity for all. Such elements can coexist with a genuine identity in commutative-identity magmas but are excluded by the inverse axiom (since a left zero cannot have an inverse).

This explains the pattern: `inv` eliminates extra idempotents (guaranteeing $left \leftrightarrow idem$), while `comm+id` alone does not. The 75% figure is not a statistical anomaly—it reflects the proportion of 3-element commutative-identity magmas that happen to lack extra idempotents.

4.7 Methodological Principle

The three-phase verification protocol—single-representative, exhaustive, saturation-confirmed—provides a complete epistemic chain:

1. **Single-representative** (fast, biased): generates hypotheses.
2. **Exhaustive** (complete, expensive): quantifies support. Reveals that 75% is not 100%.
3. **Phase 9 saturation** (mechanistic): confirms that the 75% is genuine mathematical diversity, not a derivation artifact. Modus ponens promotion does not change the count.

No single phase suffices. Phase 1 generates a false universal. Phase 2 quantifies the truth but cannot distinguish mechanism gaps from real diversity. Phase 3 closes the loop.

4.8 Corrected Findings

1. **Identity determines left–right equivalence.** Confirmed universally on all 935 models. The column is perfectly binary (all-or-nothing per class).
2. **Inverse determines identity–idempotent equivalence.** Confirmed for `id+inv` (18/18) and `assoc+comm+id+inv` (3/3). **Not** confirmed for `comm+id` (75%), `comm+id+inv` (50%), or `id` (39%). The corrected law: inverse is *sufficient* for identity–idempotent equivalence, but commutativity with identity is not.
3. **Associativity is independent.** Confirmed. Adding or removing `assoc` changes no equivalence column. This holds on both single-representative and exhaustive analyses.

5 Cross-Domain Validation

The property implication mechanism is validated on three unrelated mathematical domains, using the same infrastructure (`define_property + derive_property_implications`) with different input data.

5.1 Three Domains

5.2 Four Meta-Patterns

Four structural meta-patterns are tested across all three domains (11 instances total):

1. **Definitional equivalence in saturated structures.** In each domain, there exists a structure class where distinct-sounding properties have identical extensions: left and right identity in groups, maximal and minimal in antichains, set membership and subset-of-empty in the powerset chain. This parallels the algebraic finding in Section 4: in a saturated structure, properties that differ in definition converge in extension. All 3/3 confirmed.

Table 4: Cross-domain summary. The same mechanism produces structurally analogous findings across algebra, order theory, and set theory.

	Algebra	Order theory	Set theory
Input domain	10 axiom classes	241 labeled posets (sizes 2–4)	ZFC powerset chain
Properties	left_id, right_id, idempotent	is_maximal, is_minimal, is_isolated, is_extremal	is_set, superset_of_∅, has_member_∅, self_subset
Definitional equivalence	left ↔ right (groups)	max ↔ min (antichains)	is_set ↔ superset_of_∅
Equivalence breaks	magma: no right_id	chain: unique max ≠ min	pairing: resource boundary
Strict implication	left → idem (id class)	isolated → maximal	member_∅ → is_set
Degenerate equivalence	id ↔ idem (groups)	max ↔ isolated (antichains)	is_set ↔ self_subset

2. **Equivalence breaks in weaker structures.** When the structural conditions are relaxed, the equivalence fails. In algebra, a magma has left identity but no right identity. In order theory, a chain has a unique maximum $\neq$ minimum. In set theory, adding pairing causes a resource-dependent divergence. All 3/3 confirmed (2 structural, 1 resource).
3. **Strict implications.** In each domain, at least one implication holds without its reverse: identity implies idempotent (but not vice versa on the id class), isolated implies maximal (but not vice versa on chains), and containing empty implies being a set (but not vice versa). All 3/3 confirmed.
4. **Degenerate full equivalence.** In each domain, the most constrained structure (group, antichain, powerset chain) collapses multiple properties into full equivalence. All 3/3 confirmed.

Total: 11/11 pattern instances confirmed. No counterexample found.

5.3 Structural vs. Resource Failure

A methodologically important finding: not all equivalence failures are alike.

Definition 1. *An equivalence failure is structural if the property extensions differ at all resource levels. It is a resource failure if the extensions differ due to derivation-rate competition under a finite fact budget, but the equivalence is definitionally true.*

Diagnostic criterion. Run the engine at multiple fact-cap levels ($C_1 < C_2 < C_3$). Compute the extension gap $\delta_i = |\text{ext}(P)| - |\text{ext}(Q)|$ at each level.

- Structural: $\delta_1 = \delta_2 = \delta_3$ (constant).
- Resource: δ_i varies with C_i (cap-dependent).

On the magma (structural failure), the gap between $|\text{ext}(\text{left_id})|$ and $|\text{ext}(\text{right_id})|$ is 1 at all cap levels (100, 200, 500). On ZFC with pairing (resource failure), the gap between $|\text{ext}(\text{is_set})|$ and $|\text{ext}(\text{superset_of_empty})|$ varies: 67%, 67%, 75%.

This makes the structural/resource distinction *testable*, not just conceptual. The engine can automatically flag which non-equivalences are resource-dependent by running at multiple cap levels.

5.4 Interpretation

The cross-domain consistency is not engineered. No part of the system was designed with these specific findings in mind. The property implication mechanism was built for algebra and then applied unchanged to order theory and set theory. The four meta-patterns emerged from the data.

This suggests that the extension-inclusion mechanism captures a genuine structural invariant of mathematical reasoning: the relationship between property extensions reflects the underlying logical dependencies between the properties' definitions, regardless of the specific mathematical domain.

6 Property Composition and Concept Discovery

Sections 4–5 analyzed properties individually. This section asks: can the system *construct* new properties from existing ones, and can those constructions reveal concepts that no individual property captures?

6.1 Conjunction

`define_compose_and(name, p, q)` constructs a new property whose extension is $\text{ext}(p) \cap \text{ext}(q)$. Three rules are generated:

$$\begin{aligned} \text{has_property}(x, p) \wedge \text{has_property}(x, q) &\vdash \text{has_property}(x, \text{name}) && (\text{detect}) \\ \text{has_property}(x, \text{name}) &\vdash \text{has_property}(x, p) && (\text{split-left}) \\ \text{has_property}(x, \text{name}) &\vdash \text{has_property}(x, q) && (\text{split-right}) \end{aligned}$$

On $\mathbb{Z}_3$, $\text{two_sided_id} = \text{left_id} \wedge \text{right_id}$ has extension $\{e_0\}$, identical to both parents. Phase 6 correctly discovers $\text{two_sided_id} \leftrightarrow \text{left_id} \leftrightarrow \text{right_id}$.

On the magma (no right identity), the conjunction has empty extension, and no false equivalences are derived.

6.2 Negation

`define_negate(name, p)` constructs a property whose extension is the complement: $\text{ext}(\text{name}) = \text{domain} \setminus \text{ext}(p)$. Detection uses stratified negation at stratum 3:

$$\text{element}(x), \neg \text{has_property}(x, p) \vdash \text{has_property}(x, \text{name})$$

Negation is the key to escaping the “0 informative combinations” result of pure conjunction. Without negation, all property pairs on the test domains produce degenerate combinations (extension equals one parent, or is empty). With negation, orthogonal properties emerge.

6.3 Combination Scoring

Not all conjunctions are informative. A scoring function filters degenerate combinations:

Hard filters (score = 0):

- **Empty extension:** $\text{ext}(p \wedge q) = \emptyset$.
- **Degenerate to parent:** $\text{ext}(p \wedge q) = \text{ext}(p)$ or $\text{ext}(q)$.
- **Cross-degenerate:** $\text{ext}(p \wedge q) = \text{ext}(r)$ for some existing property r .
- **Trivial universal:** $\text{ext}(p \wedge q) = \text{domain}$.

Soft scoring (score > 0):

$$\text{score} = \underbrace{\min(1 - J_p, 1 - J_q)}_{\text{independence}} \times \underbrace{4r(1 - r)}_{\text{rarity}} \times \underbrace{1 - J(p, q)}_{\text{constraint}}$$

where $J_p = J(\text{ext}(p \wedge q), \text{ext}(p))$ is the Jaccard similarity between the combination and each parent, and $r = |\text{ext}(p \wedge q)|/|\text{domain}|$.

6.4 Results: Autonomous Concept Discovery

Without negation (pure conjunction). Across all three domains (algebra, order theory, set theory), auto-enumeration of all property pairs finds **0 informative combinations**. Every conjunction is either empty, degenerates to a parent, or equals an existing property. This is a genuine negative result: the property sets are already “minimally complete” for these structures.

With negation. On a 4-element chain $a < b < c < d$:

Table 5: Auto-discovered combinations on a 4-element chain, after negation. Cross-degenerate filter applied.

Combination	Extension	Score	Interpretation
$\neg\text{maximal} \wedge \neg\text{minimal}$	$\{b, c\}$	0.167	interior elements
<i>Filtered by cross-degenerate check:</i>			
$\text{extremal} \wedge \neg\text{minimal}$	$\{d\} = \text{ext}(\text{is_maximal})$	0	redundant
$\neg\text{maximal} \wedge \text{extremal}$	$\{a\} = \text{ext}(\text{is_minimal})$	0	redundant

The sole surviving combination, $\neg\text{maximal} \wedge \neg\text{minimal}$, identifies the *interior elements* of the chain—a recognized mathematical concept that the system discovers without being told its name or definition. The cross-degenerate filter correctly eliminates two combinations that merely reconstruct existing properties (*is_maximal* and *is_minimal*) through negation indirection.

6.5 Interpretation

1. **Negation as orthogonality generator.** Pure conjunction over a “complete” property set yields nothing new. Negation creates properties that are by construction orthogonal to existing ones, opening the combinatorial space.
2. **Cross-degenerate filtering is essential.** Without it, 2 of 3 candidates were redundant reconstructions of existing properties. The filter reduces false discovery rate from 67% to 0%.
3. **Carrier size matters.** On 3-element carriers, even negation cannot escape degeneracy (all property subsets form chains). The 4-element chain provides just enough room for a genuinely new intersection. This suggests that concept discovery requires a minimum structural complexity in the input domain.

6.6 Disjunction

`define_compose_or(name, p, q)` constructs a property with $\text{ext}(\text{name}) = \text{ext}(p) \cup \text{ext}(q)$. Only forward rules are generated (no negation-based decomposition):

$$\begin{aligned} \text{has_property}(x, p) &\vdash \text{has_property}(x, \text{name}) \\ \text{has_property}(x, q) &\vdash \text{has_property}(x, \text{name}) \end{aligned}$$

Scoring follows the same filter structure as conjunction, with soft factors adapted for unions: $\text{score} = \text{covering_penalty} \times \text{disjointness} \times \text{rarity}$.

On the 4-element chain, $\text{or}(\text{maximal}, \text{minimal}) = \{a, d\} = \text{ext}(\text{extremal})$, correctly filtered as `degenerate_to_is_extremal`.

7 Property Space Structure

The combination scoring infrastructure (Section 6) produces degenerate diagnoses for most property pairs. These “failures” are not noise—they encode the structure of the property space itself.

7.1 Degenerate Diagnosis as Evidence

Each degenerate diagnosis carries an implicit logical conclusion:

- $\text{and}(p, q)$ degenerates to $p \Rightarrow \text{ext}(p) \subseteq \text{ext}(q) \Rightarrow \text{implies}(p, q)$.
- $\text{or}(p, q)$ degenerates to $p \Rightarrow \text{ext}(q) \subseteq \text{ext}(p) \Rightarrow \text{implies}(q, p)$.
- AND degenerate ratio $> 80\%$ $\Rightarrow$ the property space is *hierarchical* (extensions form subset chains).
- OR degenerate ratio = 100% $\Rightarrow$ the property lattice is *union-closed* (no disjunction produces a new extension).

This constitutes a third independent evidence channel for property implications, alongside direct extension inclusion (Section 4) and cross-domain pattern matching (Section 5).

7.2 Three-Domain Comparison

Table 6: Property space structure across three domains. All properties include negations where applicable. “Informative” means score > 0 after all filters.

	Algebra (assoc+id, $n=3$)	Order (chain, $n=4$)	Set theory (powerset, $n=8$)
Properties	6	6	5
Pairs	15	15	10
AND degenerate	100%	93%	100%
OR degenerate	100%	100%	100%
AND informative	0	1	0
OR informative	0	0	0
Hierarchical	✓	✓	✓
Union-closed	✓	✓	✓

7.3 Meta-Level Observations

Three patterns hold across all tested domains (3 data points; low confidence, documented as observations rather than theorems):

1. **Union-closed.** In all three domains, OR produces zero informative combinations. Every disjunction either degenerates to a parent, reconstructs an existing property, or covers the entire domain. Property lattices derived from single structures appear to be closed under union.
2. **Hierarchical.** AND degenerate rates exceed 93% in all domains. Property extensions form subset chains rather than crossing freely. The sole exception— $\neg$ maximal $\wedge$ $\neg$ minimal on the 4-element chain—requires negation to create an intersection that is a proper subset of both parents.
3. **Disjunction < conjunction + negation.** Across all domains, the only informative composition operator is conjunction applied to negated properties. Disjunction adds no new concepts. This suggests that for concept discovery in derived property spaces, complementation (negation) is the essential ingredient, not aggregation (disjunction).

These observations are consistent with a general principle: when properties are derived from a single coherent structure (an algebra, an ordering, a set-theoretic construction), their extensions naturally form a lattice in which unions correspond to existing concepts. Novel concepts arise only by *subtracting* known structure (via negation), not by *adding* known structure (via disjunction).

8 Limitations and Capability Boundaries

We document the system’s limitations with the same precision as its capabilities.

8.1 Existential vs. Universal Detection

Property detection rules of the form $op(e, x, x) \vdash has_property(e, P)$ are existential: they fire when *any* witness x exists. For properties requiring universal quantification (“for all x ”), double-negation detection is needed (two strata of negation). Existential detection gives false positives on non-group structures where $op(e, x, x)$ holds for some but not all x .

8.2 ω -Rule Defense and Pure-Transfer Blocking

The ω -rule promotes saturated inductive chains to pattern facts. When a pattern fact $P(_0)$ triggers a downstream rule $P(x) \vdash Q(x)$ (pure variable transfer), the conclusion $Q(_0)$ may be incorrect. The defense blocks non-ground conclusions from pure-transfer rules. This is conservative: some correct pattern facts are blocked (e.g., $nonneg(_0)$ from $nat(_0)$).

8.3 Backward Application Disabled

Backward (apply) rules $has_property(e, P) \vdash formula(e, \dots)$ cause cascading pattern facts in algebraic contexts where the operation relation has rich structure. Currently, property detection is forward-only. Backward inference from properties to formulas is deferred until proper grounding guards are designed.

8.4 Property Implication: Observed vs. Deductive

$\text{implies_observed}(P, Q)$ is computed from current extensions: $\text{ext}(P) \subseteq \text{ext}(Q)$. At fixpoint, this is a theorem of the structure (all derivable instances have been derived). The engine promotes to deductive $\text{implies}(P, Q)$, enabling modus ponens.

However, single-representative sampling can produce false universals: a property equivalence that holds on one representative may fail on other models in the same axiom class. Exhaustive verification (Section 4.4) is required to distinguish universal from partial equivalences. The three-phase protocol (sample $\rightarrow$ exhaust $\rightarrow$ saturate) provides the complete epistemic chain.

8.5 Resource-Dependent Equivalence Failure

When derivation rules have different growth rates (e.g., pairing $O(n^2)$ vs. subset $O(n)$), a finite fact budget is exhausted before slower rules catch up. Definitionally true equivalences appear as non-equivalences. The temporal convergence diagnostic (varying the fact cap) can detect this, but cannot resolve it without increasing the budget or removing competing rules.

8.6 Carrier Size

All algebraic results are on 3-element carriers. Transfer to $\mathbb{Z}_4$ is verified (6/6 predictions correct), but systematic exploration at $n > 3$ is computationally expensive ($4^{16} \approx 4.3 \times 10^9$ operations at $n = 4$).

8.7 Zorn’s Lemma and Second-Order Reasoning

The system can verify properties of specific finite posets (maximal element existence) and can construct proof artifacts (chains, termination witnesses). It cannot prove universal theorems about *all* posets, which requires structural induction (second-order reasoning). Zorn’s Lemma is a capability boundary marker, not a system target. Depth-bounded “maximal” elements are artifacts of the depth limit, not genuine mathematical maximality; proof by contradiction correctly distinguishes the two.

This limitation is not specific to Relatum. Any finitary syntactic reasoning system operating on first-order Horn clauses with stratified negation faces the same boundary: universal theorems about infinite families of structures require meta-level induction that cannot be expressed within the object-level closure. This is a consequence of the expressive gap between first-order and second-order logic, not an engineering deficiency.

9 Conclusion

We have presented Relatum, a relation-first reasoning engine that discovers mathematical structure through two independent mechanisms—inductive closure scoring and property implication analysis—and demonstrated their cross-validation across three mathematical domains.

9.1 Summary of Results

- **Algebraic discovery:** from 19,683 raw operations, the system identifies abelian groups as the optimal axiom class through dual-signal alignment, with formal equivalence to standard group axioms and successful transfer to $\mathbb{Z}_4$.
- **Property implication:** across 10 axiom classes and 935 models, the system discovers that identity determines left–right equivalence (935/935), inverse guarantees identity–idempotent equivalence (fully verified on **id+inv**), but commutativity with identity yields

only 75% equivalence—revealing extra-idempotent diversity within axiom classes. A three-phase verification protocol (single-representative $\rightarrow$ exhaustive $\rightarrow$ saturation-confirmed) establishes this as genuine mathematical diversity, not a mechanism artifact.

- **Cross-domain generality:** the property mechanism produces structurally analogous findings on algebra (10 classes), order theory (241 posets), and set theory (ZFC). Four meta-patterns—definitional equivalence, equivalence breaking, strict implication, degenerate collapse—repeat across all three domains (11/11 confirmed).
- **Failure diagnosis:** the system distinguishes structural failure (genuine mathematical non-equivalence) from resource failure (derivation-rate competition) by varying the fact budget. This makes the limitation not just acknowledged but testable.
- **Concept discovery:** property conjunction without negation yields 0 informative combinations across all domains (properties are already “minimally complete”). Adding negation as an orthogonality generator produces genuinely new concepts: “interior elements” on a 4-element chain. Cross-degenerate filtering eliminates 67% of false discoveries.

9.2 Central Claim

The structural boundary between groups and weaker algebraic structures is independently discoverable by two mechanisms that share no intermediate results. The property implication mechanism generalizes to non-algebraic domains without modification. These findings suggest that extension-based property analysis captures a domain-independent structural invariant of mathematical reasoning.

9.3 Future Directions

- **Deeper composition:** recursive composition ($p \wedge (q \wedge r)$), disjunction ($p \vee q$), and conditional properties ($p \mid q$) as additional constructors.
- **ω -rule on property implications:** promoting observed cross-structure implications to meta-level universal claims, with appropriate defenses against the risks identified in Section 8.
- **Larger carriers:** extending the algebraic enumeration beyond $n = 3$, potentially using symmetry reduction to manage the n^{n^2} growth.
- **Proto-level axiom discovery:** the beam search with compression + purity scoring autonomously discovers transitivity from raw binary relation data. Extending this to discover full axiom systems (e.g., lattice axioms from poset data) is a natural next step.